Evaluating estimators

- Parametric model $\mathcal{F} = \{f(\cdot;\theta) : \theta \in \Theta \subseteq \mathbb{R}^d\}$
- $X_1, \ldots, X_n$ random sample
- $T(X_1, \ldots, X_n)$ be an estimator $\psi(\theta)$

Def: The bias of an estimator $T(X_1, \ldots, X_n)$ of $\psi(\theta)$ is

$$\text{bias}(T; \psi(\theta)) = \mathbb{E}_\theta\, T(X_1, \ldots, X_n) - \psi(\theta)$$

(CB Def 7.3.2)

An estimator is unbiased if the bias is zero for all $\theta \in \Theta$.

Example: If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, then $\bar{X}$ is an unbiased estimator of μ. Is $\hat{\sigma}^2 = n^{-1}\sum_{j=1}^{n}(X_j - \bar{X})^2$ an unbiased estimator of σ^2?

Note $\hat{\sigma}^2 = \frac{n-1}{n} S^2$. We saw that $\frac{(n-1)}{\sigma^2} S^2 \sim \chi^2_{n-1}$.

$$\mathbb{E}\left(\frac{(n-1)}{\sigma^2} S^2\right) = \frac{(n-1)}{\sigma^2}\mathbb{E}(S^2) = n-1 \Rightarrow \mathbb{E}(S^2) = \sigma^2.$$

Then

$$\mathbb{E}(\hat{\sigma}^2) = \mathbb{E}\left(\frac{n-1}{n} S^2\right) = \frac{n-1}{n} \cdot \mathbb{E}(S^2) = \frac{n-1}{n}\sigma^2$$

The bias of $\hat{\sigma}^2$ is $\text{bias}(\hat{\sigma}^2; \sigma^2) = \mathbb{E}_\theta(\hat{\sigma}^2) - \sigma^2 = -\frac{1}{n}\sigma^2$.

Note this holds without the normality assumptions.

Example (Normal) If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, then X_1 is an unbiased estimator of μ. Is it a good estimator?

Def: The mean squared error (MSE) of an estimator $T(X_1, \ldots, X_n)$ of $\psi(\theta)$ is

$$\text{MSE}(T; \psi(\theta)) = \mathbb{E}_\theta\left[(T(X_1, \ldots, X_n) - \psi(\theta))^2\right]$$

(CB Def 7.3.1)

Decomposition of MSE

$$MSE(T;\psi(\theta)) = E_\theta\left[\left(T(X_1,\dots,X_n) - E_\theta T(X_1,\dots,X_n) + E_\theta T(X_1,\dots,X_n) - \psi(\theta)\right)^2\right]$$

$$= E_\theta\left[\left(T(X_1,\dots,X_n) - E_\theta T(X_1,\dots,X_n)\right)^2\right] \leftarrow \text{Variance of } T(X_1,\dots,X_n)$$

$$+2\left(E_\theta T(X_1,\dots,X_n) - \psi(\theta)\right) E_\theta\left[T(X_1,\dots,X_n) - E_\theta T(X_1,\dots,X_n)\right] = 0$$

$$+\left(E_\theta T(X_1,\dots,X_n) - \psi(\theta)\right)^2 \leftarrow \text{bias squared}$$

$$= \text{Var}_\theta T(X_1,\dots,X_n) + \text{bias}(T;\psi(\theta))^2.$$

Example (Normal) The MSE of $\bar{X}$ as an estimator of μ is

$$MSE(\bar{X};\mu) = \text{Var}_\theta \bar{X} = \frac{\sigma^2}{n}.$$

since $\bar{X}$ is an unbiased estimator of μ.

Example (Normal) Which estimator of σ^2 has the smaller MSE, S^2 or $\hat{\sigma}^2 = \frac{n-1}{n} S^2$?

Since S^2 is an unbiased estimator of σ^2, so $MSE(S^2;\sigma^2) = \text{Var}_\theta(S^2)$. As $\frac{n-1}{\sigma^2} S^2 \sim \chi^2_{n-1}$,

$$\text{Var}_\theta\left(\frac{n-1}{\sigma^2} S^2\right) = \frac{(n-1)^2}{\sigma^4} \text{Var}_\theta(S^2) = 2(n-1)$$

$$\Rightarrow \text{Var}_\theta(S^2) = \frac{2\sigma^4}{(n-1)}$$

$$\text{Var}_\theta(\hat{\sigma}^2) = \text{Var}_\theta\left(\frac{n-1}{n} S^2\right) = \frac{(n-1)^2}{n^2} \text{Var}_\theta(S^2) = \frac{2(n-1)\sigma^4}{n^2}$$

$$MSE(\hat{\sigma}^2;\sigma^2) = \frac{2(n-1)\sigma^4}{n^2} + \frac{1}{n^2}\sigma^4 = \frac{\sigma^4(2n-1)}{n^2} < \frac{2\sigma^4}{(n-1)}$$

So $\hat{\sigma}^2$ has a smaller MSE than S^2

Def: An estimator $T_n(X_1,\dots,X_n)$ of $\psi(\theta)$ is said to be (weakly) consistent if for all $\varepsilon>0$

$$\lim_{n\to\infty} \mathbb{P}_\theta\left(|T_n(X_1,\dots,X_n)-\psi(\theta)|>\varepsilon\right)=0$$

(CB Def 10.1.1)

We can write $T_n(X_1,\dots,X_n) \xrightarrow{p} \psi(\theta)$.

(convergence in probability)

Recall the weak law of large numbers (STAT2003 – claim 55)

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}-\mu|>\varepsilon)=0$$ for a random sample $X_1,\dots,X_n$ with finite mean.

A simple check for consistency is MSE $\to 0$.

Markov's inequality: For a positive random variable W and $c>0$,

$$\mathbb{P}(W>c) \le \frac{\mathbb{E}W}{c}$$

Apply Markov's inequality to $|T_n(X_1,\dots,X_n)-\psi(\theta)|^2=W$ and take $c=\varepsilon^2$.

$$\mathbb{P}_\theta\left(|T_n(X_1,\dots,X_n)-\psi(\theta)|^2>\varepsilon^2\right) \le \frac{MSE(T;\psi(\theta))}{\varepsilon^2}$$

So if MSE $\to 0$, then $T_n(X_1,\dots,X_n) \xrightarrow{p} \psi(\theta)$.

Example (Bernoulli) $X_1,\dots,X_n \sim$ Bernoulli(p) - The estimator $\hat{p}=\frac{1}{n}\sum_{j=1}^{n} X_j$ is consistent.

Homework check $\mathbb{E}_p(\hat{p})=p$ and $Var_p(\hat{p})=\frac{p(1-p)}{n}$

$$MSE(\hat{p};p)=\frac{p(1-p)}{n} \to 0 \text{ as } n\to\infty \text{ for all } p\in[0,1]$$

Hence, $\hat{p}$ is a consistent estimator of p.

Theorem: If $W_1, W_2, \ldots$ converges in probability to W and h is a continuous function, then $h(W_1), h(W_2), \ldots$ converges in probability to $h(W)$.

Example: (Bernoulli) Since $\hat{p}$ is a consistent estimator of p, ~~$(1-\hat{p})/\hat{p}$~~ $\dfrac{\hat{p}}{1-\hat{p}}$ is consistent estimator of odds.